Analyzing and Forecasting Air Quality

COHNDDS242F-009 – H.M.N.B.Herath

Research Question

How to analyze the historical trends and patterns in PM2.5 levels in India and how to apply univariate time series models to accurately forecast future PM2.5 levels ?

Research Objective

To analyze the historical trends and patterns in PM2.5 levels in India and apply univariate time series models to accurately forecast future PM2.5 levels.

Research Gap

There are only a few research that focus on only PM2.5, and none of the research has used time series for analysis and forecasting.

Literature Review

Time Series Analysis of Air Quality in India:

Analyzed the impact of PM 2.5, PM 10, seasons, and COVID-19 on AQI, using Prophet for accurate forecasting.

Air Quality Analysis in India:

Studied the effects of industrial, vehicular, and household pollutants on air quality, and predicted future AQI using time-series algorithms.

Data Collection Methods

Obtained the Air_Quality_India dataset from Kaggle (secondary data).

Data Preprocessing

Checking the missing values

```
colSums(is.na(Data))
```

```
## Timestamp      Year     Month       Day      Hour     PM2.5
##         0         0         0         0         0         0
```

Checking the outliers and replacing outliers

```
# Ensuring the column PM2.5 exists in the data
df <- Data_df[!is.na(Data_df$PM2.5), ]

# Calculate the IQR and quantile bounds
Q1 <- quantile(Data$PM2.5, 0.25, na.rm = TRUE)
Q3 <- quantile(Data$PM2.5, 0.75, na.rm = TRUE)
IQR_value <- Q3 - Q1

lower_bound <- Q1 - 1.5 * IQR_value
upper_bound <- Q3 + 1.5 * IQR_value

# Compute the median value
median_value <- median(Data$PM2.5, na.rm = TRUE)

# Replace outliers with the median value
Data_df <- Data %>%
  mutate(PM2.5 = ifelse(PM2.5 < lower_bound | PM2.5 > upper_bound, median_value, PM2.5))
} else {
  stop("Column 'PM2.5' not found in the dataset!")
}

# Display the first few rows
head(Data_df)
```

```
##             Timestamp Year Month Day Hour  PM2.5
## 0 11/7/2017 12:00 2017   11   7  12  64.55
## 1 11/7/2017 13:00 2017   11   7  13  63.75
## 2 11/7/2017 14:00 2017   11   7  14  90.36
## 3 11/7/2017 15:00 2017   11   7  15 109.66
## 5 11/7/2017 16:00 2017   11   7  16 116.50
## 6 11/7/2017 17:00 2017   11   7  17  45.75
```

Convert to time series format, Aggregate data to monthly level

```
# Convert to time series format
PM2.5_ts <- ts(Data_df$PM2.5,
               start = c(as.numeric(substr(min(Data_df$year_month), 1, 4)),
                         as.numeric(substr(min(Data_df$year_month), 6, 7))),
               frequency = 12)
```

```
# Convert the 'Timestamp' column to Date format
Data_df$Date <- as.Date(Data_df$Timestamp, format = "%m/%d/%Y %H:%M")
str(Data_df$Date)
```

```
##  Date[1:36192], format: "2017-11-07" "2017-11-07" "2017-11-07" "2017-11-07"
```

```
# Aggregate data to the monthly level
library(dplyr)

Data_df <- Data_df %>%
  mutate(year_month = format(Date, "%Y-%m")) %>%
  group_by(year_month) %>%
  summarise(PM2.5 = mean(PM2.5, na.rm = TRUE))

# Display the first few rows
head(Data_df)
```

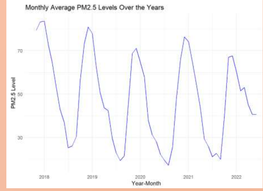
```
## # A tibble: 6 × 2
##   year_month PM2.5
##   <chr>      <dbl>
## 1 2017-11     79.5
## 2 2017-12     83.3
## 3 2018-01     83.7
## 4 2018-02     72.5
## 5 2018-03     64.8
## 6 2018-04     53.2
```

Applied Log Transformation

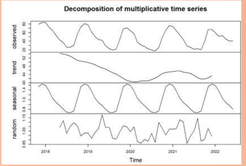
```
#Log transformation
log <- log(Data_df$PM2.5)
```

Exploratory Data Analysis (EDA)

This plot shows yearly seasonality and decreasing trend from 2018 – 2020 in PM2.5 level

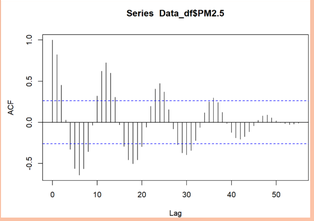


This shows strong seasonal pattern and declining and recovering trend and also minor random fluctuation



The decomposition plot shows the down trend and also shows seasonal pattern after removing the trend and pattern you can see the random fluctuation in the plot

The Autocorrelation plot shows how past values correlated with present values. This shows seasonality over time.



Mean and Variance

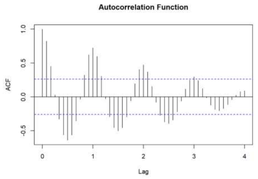
```
# Display statistics
print(paste("Mean:", mean_pm25))

## [1] "Mean: 47.8428181408359"

print(paste("Variance:", var_pm25))

## [1] "Variance: 411.121117394095"
```

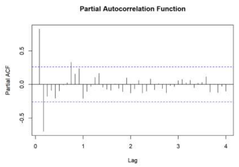
Stationarity and Transformation

Conducted the ADF test to see If the series is stationary or not, and it came as p value is 0.01 which means the series is stationary

```
##
##  Augmented Dickey-Fuller Test
##
## data:  PM2.5_ts
## Dickey-Fuller = -4.869, Lag order = 3, p-value = 0.01
## alternative hypothesis: stationary
```

The ACF plot shows strong correlation in initial lags indicating the non - stationary while PCAF suggest AR structure. And the ACF plot show seasonality over the time.





Modeling

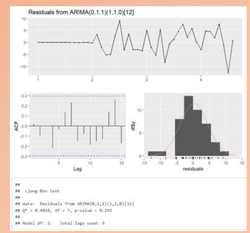
Conduct time series modeling to this series by splitting the data and applying auto.arima to find SARIMA model and ETS model and used cross-validation to select the most accurate model for the series.

```
# Split the dataset into training and testing data
train_size <- floor(0.8 * length(PM2.5_ts))
train_data <- PM2.5_ts[1:train_size]
test_data <- PM2.5_ts[(train_size + 1):length(PM2.5_ts)]

# Check the split
length(train_data)  # Should be 44 months (80%)
```

Fit SARIMA model

```
# Automatically fit SARIMA model
sarima_model <- auto.arima(train_ts, seasonal = TRUE)

# Print model summary
summary(sarima_model)

## Series: train_ts
## ARIMA(0,1,1)(1,1,0)[12]
##
## Coefficients:
##          ma1     sar1
##      -0.1377  -0.3051
## s.e.  0.1933   0.2825
##
## sigma^2 = 26.43:  log likelihood = -94.85
## AIC=195.7   AICc=196.59   BIC=200
##
## Training set error measures:
##               ME     RMSE      MAE      MPE     MAPE     MASE
## Training set 0.2826582 4.189759 2.545201 0.9628662 8.029742 0.4012517
##                    ACF1
## Training set 0.02051815
```

Fitting the SARIMA Model using auto.arima because I identify seasonality in ACF plot and The summary of the SARIMA model shows lower MAPE, RMSE, and MAE values.

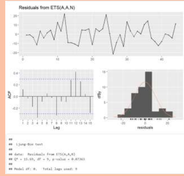


As the residuals shows the P value is higher than 0.05 which suggests that this model does not have significant autocorrelation and the spikes in ACF plot are within the blue line and the fluctuations shows no clear pattern, that means the SARIMA model capture all the patterns and trends in the series and it's a good model.

Fit ETS model

```
ets_model <- ets(train_data)
summary(ets_model)

## ETS(A,A,N)
##
## Call:
## ets(y = train_data)
##
##  Smoothing parameters:
##    alpha = 0.0001
##    beta  = 0.0001
##
##  Initial states:
##    l = 79.4853
##    b = 0.5195
##
##  sigma:  9.97
##
##     AIC     AICc      BIC
## 371.3833 373.5444 380.0464
##
## Training set error measures:
##                   ME     RMSE      MAE      MPE     MAPE     MASE
## Training set 0.199503 9.220015 7.24771  3.492546 17.28184 0.788639
```

Fit the ETS model to check it if ETS model fit for series.




```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(A,A,N)
## Q* = 15.03, df = 5, p-value = 0.01761
##
## Model df: 0.   Total lags used: 5
```

The plot shows the fluctuations around zero with no clear pattern and the ACF plot shows the all the spikes are within the blue line and also the p value is higher than 0.05. these all shows that this model is good model

Cross - Validation

After fitting the SARIMA and ETS models and checking the residuals of both models, I applied cross-validation to find the best model the series.

```
##                    ME     RMSE      MAE      MPE     MAPE      MASE
## Training set  0.28265823 4.189759 2.545201 0.9628662  8.029742 0.3209050
## Test set     -0.04704323 8.196055 6.902789 1.0077413 16.582014 0.7511074
##                   ACF1
## Training set 0.02051815
## Test set             NA
```

```
ets_accuracy
```

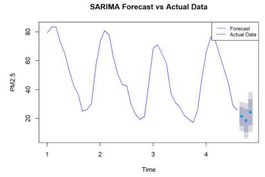
```
##                    ME     RMSE      MAE      MPE     MAPE      MASE     ACF1
## Training set -0.199503  9.220015  7.24771  3.492546 17.28184 0.788639 0.1203609
## Test set     40.738183 47.138808 40.95250 81.800686 82.81774 4.456131       NA
```

```
#as the results of accuracy shows the MAPE value in SARIMA model is lower than the MAPE value in ets model which suggest tha
t Sarima model is the best fit for the dataset.
```

Applied the cross – validation and according to the cross- validation it shows that SARIMA model's MAPE and MAE and RMSE values are lower than ETS model which suggest the SARIMA model is the best fitting model for this series. This shows the SARIMA model is best fit for both training and test data. Overall SARIMA model is the best fit for the series

Forecasting

```
library(forecast)

# Generate forecasts for the next 3 months
sarima_forecast <- forecast(sarima_model, h = 3)

# Plot the forecast
plot(sarima_forecast, main = "SARIMA Forecast vs Actual Data", xlab = "Time", ylab = "PM2.5", col = "blue")

# Add a legend to the plot
legend("topright", legend = c("Forecast", "Actual Data"), col = c("blue", "darkgreen"), lty = 1, cex = 0.8)
```



After doing the cross-validation and finding the best fitting model, it shows that SARIMA is the best fitting model so Using the SARIMA model I have forecasted the PM2.5 values for next 3 months and the plot shows the actual PM2.5 value and the Forecast PM2.5 values for next 3 months.

Results and Conclusions

The SARIMA model was used to forecast future PM2.5 values and the Model fit best with series with lower MAPE , RMSE and MAE values.

Strengths

This study covers all the aspects of time series forecasting and applies both the SARIMA and ETS models and validates them using cross-validation.

Limitations

Only use the Air_Quality_India dataset from Kaggle, which didn't capture all the all the variations in PM2.5 levels and only focuses on univariate time series forecasting.

Recommendations

Expand the data sources, like use other different sources that can get data and adopt multivariate models and factors for this study.